An unconditional Lean proof of the Erdős 690 classification

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Abstract

A sharper symmetric-sum comparison, obtained from a finite Bernoulli mean–mode theorem, combines with elementary Chebyshev estimates to give an explicit uniform argument for every $k \geq 38001$. It also removes the prime-counting premise from the existing range $49 \leq k \leq 38000$. The argument uses no zeta-zero verification, no new prime enumeration, and no enlarged finite certificate range. The already certified fixed gap and twin-prime witnesses and the closed proofs for $k \leq 48$ are retained. The elementary estimates, infinite tail, and exact all-k assembly have been implemented and compiled in Lean 4.34.0 without an analytic hypothesis.

Scope. This removes the additional analytic verification while retaining the original fixed witnesses. This paper describes the current unconditional source in this repository. The previous conditional submission [7] is superseded by the new entry point `Solution.lean`. No new prime or zero certificate is introduced.

Provenance and review. The density formula and initial cases follow Cambie [1]; the complete classification and descent-before-ascent strategy follow Wang and Crapis [2]. This work supplies an elementary adaptation and its AI-assisted Lean implementation, without claiming priority for the classification. The author directed the development using Codex. Independent expert review and independent replay of the updated full proof are not claimed. The project-authored paper is available under Apache-2.0 or CC-BY-4.0, at the recipient's option.

The density being classified. For a finite list L, let $F(L, r)$ be the coefficient of z^r in $\prod_{p \in L}((p-1)/p + z/p)$. The exact rational density at a prime p is $F(\{q : q < p,\ q \text{ prime}\}, k-1)/p$. Periodic counting proves that it is the natural density of positive integers whose kth smallest distinct prime divisor is p. The corresponding Lean theorem is `hasNaturalDensity_primeFactorDensity`.

1 The coefficient comparison

Let $p_1 = 2, p_2 = 3, \ldots, r = k-1$, and $g_i = p_{i+1} - p_i$. For a finite prime prefix $\mathcal{P}$, write e_j for the elementary symmetric sum in weights $w_p = 1/(p-1)$ and $R_r = e_{r-1}/e_r$. The exact density identity in the existing proof is

$$d_{r+1}(p_{i+1}) > d_{r+1}(p_i) \quad \Longleftrightarrow \quad R_r(\{p : p < p_i,\ p \text{ prime}\}) > g_i + 1,$$

when $e_r > 0$. Reversing the strict inequality gives a descent. An earlier strict descent and a later strict ascent exclude unimodality. Put

$$A(x) = \sum_{p \leq x} \frac{1}{p-1}, \qquad B(x) = \sum_{p \leq x} \frac{\log p}{p}, \qquad E(x) = B(x) - \log x.$$

Every prime sum displayed has the indicated finite support.

Lemma 1. *Let $1 \le r$ be an integer. For any finite prime prefix and $t > 0$, put $\mu(t) = \sum_p t/(p-1+t)$. If $\mu(t) > r$, then $e_r > 0$ and $R_r < t$. If $e_r > 0$, then $R_r \ge r/A$, where $A = \sum_p w_p$.*

Proof. Put $a_j = tw_{p_j}$ and let $c_m = e_m(a_1, \dots, a_n)$, zero outside $0 \le m \le n$. Since every $a_j/(1 + a_j) < 1$, the hypothesis implies $n > r$, so $c_r > 0$. Suppose $c_{r-1} \ge c_r$. Delete the jth variable and call the resulting coefficients b_m. Newton's inequalities give $b_{r-1}^2 \ge b_{r-2}b_r$. If $b_r > b_{r-1}$, then $b_{r-2} < b_{r-1}$ and

$$c_{r-1} = b_{r-1} + a_j b_{r-2} < b_r + a_j b_{r-1} = c_r,$$

a contradiction. Thus $b_r \le b_{r-1}$ and $c_r \le (1 + a_j)b_{r-1}$. Summing $a_j c_r/(1 + a_j) \le a_j b_{r-1}$ gives $\mu(t)c_r \le rc_r$, again a contradiction. Consequently $c_{r-1} < c_r$, or $e_{r-1} < te_r$. For the lower comparison,

$$re_r = \sum_{|S|=r-1} w_S \sum_{p \notin S} w_p \le Ae_{r-1}.$$

□

The needed log-concavity and deletion comparison are proved in the Lean implementation. Darroch's broader finite mean–mode theorem [3, 4] provides related context; the strict comparison above is sufficient here.

2 Elementary estimates and their domains

We use the classical elementary Chebyshev inequalities, not an explicit PNT or finite-height RH theorem. For $x \ge 900$,

$$\psi(x) \ge ax - \tfrac{5}{2}\log x + 1, \tag{1}$$

$$\vartheta(x) \le \tfrac{6}{5}ax - a\sqrt{x} + \frac{5}{4\log 6}(\log x)^2 + \tfrac{5}{2}\log x, \tag{2}$$

where $a = \log(2^{1/2}3^{1/3}5^{1/5}30^{-1/30})$ and $0.92129 \le a \le 0.92130$; see [5], Section 1.1. Also use the elementary bounds $\vartheta(x) < (\log 4)x$ and $\psi(x) - \vartheta(x) \le 2\sqrt{x}\log x$.

Lemma 2. *One has*

$$\vartheta(x) \le 1.16x \ (x \ge 1000), \qquad \vartheta(x) \le 1.12x \ (x \ge 2000), \qquad \vartheta(x) \ge 0.4x \ (x \ge 1000),$$

and $\vartheta(x) \ge 0.92x$ for $x \ge 2^{30}$.

Proof. Let $Q(v) = 0.7v^2 + 2.5v + 2$. Equation (2) gives

$$\vartheta(x)/x < 1.10556 - 0.9212/\sqrt{x} + Q(\log x)/x.$$

The function $Q(v)e^{-v}$ decreases for $v \ge \log 1000$. Thus dropping the negative term and using $\log 1000 < 7$ gives $1.10556 + Q(7)/1000 = 1.15936 < 1.16$.

For $v \ge \log 2000$, $Q(v)e^{-v/2}$ decreases: its derivative has the sign of $-0.35v^2 + 0.15v + 1.5 < 0$. While $Q(\log x)/\sqrt{x} - 0.9212$ is positive, multiplying it by $1/\sqrt{x}$ preserves this decrease; afterwards the correction is nonpositive. Hence its positive part is bounded by its value at 2000. Using $\log 2000 < 7.61$ and $\sqrt{2000} < 45$ gives

$$\vartheta(x)/x < 1.10556 + Q(7.61)/2000 - 0.9212/45 < 1.12.$$

For the lower bounds combine (1) with $\psi - \vartheta \le 2\sqrt{x}\log x$. At 1000, use $\log 1000 < 7$, $\sqrt{1000} > 31$, and monotonicity of $(\log x)/x$ and $(\log x)/\sqrt{x}$. For $x \ge 2^{30}$, the two loss terms decrease. At the endpoint use $\log(2^{30}) < 21$ and $\sqrt{2^{30}} = 32768$:

$$\frac{2.5(21)+1}{2^{30}} + \frac{42}{32768} < 0.00129.$$

Together with $a \ge 0.92129$, this gives the lower bound $0.92x$. □

Lemma 3. *For every real $x \ge 1000$, $-2 < E(x) < 1/5$.*

Proof. Let $n = \lfloor x \rfloor$. Factorial identities and integral bounds give

$$B(n) \le \log(n!)/n + \vartheta(n)/n \le \log n + 0.16 + (1+\log n)/n < \log n + 1/5.$$

Here $(1+\log n)/n < 8/1000$. For a lower bound put $D = \sum_p \log p/[p(p-1)]$. The six primes through 13 contribute less than 0.7, and

$$\sum_{m=14}^{\infty} \frac{\log m}{m(m-1)} \le \frac{\log 14 + 1}{13} < 0.28.$$

The last inequality follows by telescoping summation by parts and $\log(1+1/m) \le 1/m$. Thus $D < 0.98$. Since $\log(n!)/n \le B(n) + D$ and $\log(n!)/n \ge \log n - 1$, $B(n) > \log n - 1.98$. Use $B(x) = B(n)$ and $\log x - \log n < 1/n$. □

Lemma 4. *For $x \ge 1000$,*

$$A(x) < \log\log x + 1.1 + \frac{4}{\log x}.$$

Proof. Elementary Mertens summation gives

$$\sum_{p \le x} \frac{1}{p} - \log\log x - M = \frac{E(x)}{\log x} - \int_x^{\infty} \frac{E(u)}{u(\log u)^2}\, du.$$

Lemma 3 bounds its absolute value by $2.2/\log x$. The elementary identity $M = \gamma + \sum_p (\log(1-1/p) + 1/p)$ identifies the constant [6]; no zeta-zero statement is involved. The bound $\gamma \le H_{50} - \log 50 < 0.59$, followed by the five primes $2, 3, 5, 7, 11$, yields

$$M < 0.59 + \log(16/77) + 2927/2310 < 0.29.$$

For $C = \sum_p 1/[p(p-1)]$, sum the nine primes through 23. For the rest, use

$$\sum_{m=24}^{\infty} \frac{1}{m(m-1)} \le \frac{24}{23} \sum_{m=24}^{\infty} \frac{1}{m^2} \le \frac{24}{529}.$$

The resulting exact rational bound is $C < 0.81$. Since $A(x) \le \sum_{p \le x} 1/p + C$, the constants sum to 1.1. Finally relax $2.2/\log x$ to $4/\log x$. □

3 A smoothed lower bound retaining the small-variable contribution

Lemma 5. *For $X \ge t \ge 4752$,*

$$\sum_{p\le X} \frac{1}{p+t} \ge \log\log X - \log\log t - \frac{\log(1+1000/t)}{\log t} - \frac{11/5}{\log t + 2}. \tag{S}$$

Proof. Set $Y = 1000$ and $f(u) = u/((u+t)\log u)$. Summation by parts on the exact interval $(Y, X]$ gives

$$\sum_{Y<p\le X} \frac{1}{p+t} = \int_Y^X \frac{du}{(u+t)\log u} + f(X)E(X) - f(Y)E(Y) - \int_Y^X E(u)f'(u)\,du.$$

The function f increases and then decreases on $[Y,\infty)$: f' has the sign of $t(\log u - 1) - u$, positive at Y with exactly one later zero. Write $F = \sup_{u\ge Y} f(u)$. Using $-2 < E < 1/5$, split the last integral at the maximum. The error is at least $-(11/5)F + (1/5 - E(Y))f(Y) \ge -(11/5)F$. If the maximum is after X, replace F in this calculation by $f(X)$.

For $z = t/u$, the inequality $\log(z\log u) \le z\log u - 1$ and $\log\log u > 1$ show $\log t + 2 \le (1 + t/u)\log u$. Therefore $F \le 1/(\log t + 2)$. Finally,

$$\int_Y^t \frac{du}{(u+t)\log u} \ge \frac{\log(2t/(Y+t))}{\log t}, \qquad \int_t^X \frac{du}{(u+t)\log u} \ge \log\log X - \log\log t - \frac{\log 2}{\log t}.$$

The $\log 2$ contributions cancel. Omitted primes $p \le Y$ contribute positively, proving (S). □

For a prefix containing every prime through X,

$$\mu(t) \ge t \sum_{p\le X} \frac{1}{p+t}. \tag{T}$$

This follows from $p - 1 + t < p + t$. Together with Lemma 1, (S) and (T) give a complete descent estimate.

4 Symbolic primorial selection

For real $r \ge 38000$ define

$$v = \log r, \quad H = v - \log v + 6/5, \quad \ell = \frac{r}{(11/10)H}, \quad U = \log(5\ell), \quad t = \frac{25}{14}(\ell - U). \tag{P}$$

For integer r, choose the first primorial P with $\log P \ge \ell$. Let $q^- < q$ be its largest two prime factors, and $L = \log P$. This is an existence construction; no primorial or prime list is evaluated.

Lemma 6. *One has $\ell \ge 3670$, $U/\ell < 0.0027$, $t > 6535$, and*

$$\ell \le L < \ell + U, \qquad 2q^- > t.$$

Proof. The function $r/H(\log r)$ increases. At $r = 38000$, $H < 9.4$, so $\ell \ge 38000/(1.1\cdot 9.4) > 3670$. Monotonicity of $\log(5x)/x$, with $\log 18350 < 9.82$, gives $U/\ell < 0.0027$ and $t > (25/14)(3670 - 9.82) > 6535$.

Minimality gives $\vartheta(q^-) < \ell$. If $q^- \ge 1000$, Lemma 2 gives $q^- < 2.5\ell$; otherwise this is automatic. Bertrand's theorem gives $q < 2q^- < 5\ell$. Hence $\log q < U$ and $\ell \le L < \ell + U$. Moreover $\vartheta(q^-) = L - \log q > \ell - U > 3660$. The rough upper bound $\vartheta(q^-) < 1.4q^-$ implies $q^- > 2000$. Apply $\vartheta(q^-) \le 1.12q^-$ to obtain $2q^- > (25/14)(\ell - U) = t$. □

5 The entire scalar ray, proved by differentiation

Lemma 7. *For every real $r \geq 38000$,*

$$t\left(\log \ell - \log\log t - \frac{\log(1+1000/t)}{\log t} - \frac{11/5}{\log t + 2}\right) > \frac{2001}{2000} r. \tag{D}$$

With $M_* = \ell + \log(2000\ell)$, one also has

$$(1.09M_* + 2)(\log M_* + 1.1 + 4/M_*) < r. \tag{A}$$

Proof. Let $v_0 = 2109/200 = 10.545$. For $v \geq v_0$, $9 < H < v$ and $H/v \geq 0.89$; the derivative of H/v is $(\log v - 2.2)/v^2 > 0$, and its value at v_0 exceeds 0.89.

By Lemma 6, $\log t > 8.78$ and $1000/t < 0.154$. Also $\log t < v - \log H + 0.485$, since $\log(125/77) < 0.485$. Use $\log(1+s) \leq s$, $\log 1.1 < 0.096$, and

$$0.096 + 0.154/8.78 + 2.2/10.78 < 0.319.$$

The bracket in (D) consequently exceeds $B_0(v) = v - \log H - \log(v - \log H + 0.485) - 0.319$. Put $J = v - \log H + 0.485$. Then $J > 8$, $0 < H' < 1$, $H'/H < 1/9$, and $0 < J' < 1$. Thus

$$(B_0 - 0.618H)' > 1 - 1/9 - 1/8 - 0.618 > 0.$$

At v_0, $9.3 < H < 9.39$, $2.23 < \log H < 2.24$, and $\log J < \log 8.8 < 2.175$, so

$$B_0(v_0) - 0.618H(v_0) > 10.545 - 2.24 - 2.175 - 0.319 - 0.618(9.39) = 0.00798 > 0.$$

Hence the bracket exceeds $0.618H$ on the whole infinite ray. Since $t/r = (125/77)(1 - U/\ell)/H$,

$$\frac{t}{r}(0.618H) > (125/77)(0.9973)(0.618) = 1.000537987\ldots > 2001/2000.$$

This proves (D) by calculus, not by scanning integer parameters.

For (A), monotonicity of $\log(2000x)/x$ and $\log 7340000 < 15.81$ give $M_* < 1.005\ell$. Since $1.1H/v \geq 0.979$ and $\log 0.979 > -0.022$,

$$\log \ell + 1.1 = H - 0.1 - \log(1.1H/v) < H - 0.078.$$

It follows that $\log M_* + 1.1 + 4/M_* < H - 0.078 + 0.005 + 4/3670 < H$. Also $1.09(1.005) + 2/3670 < 1.1$, so $1.09M_* + 2 < 1.1\ell$. Their product is less than $1.1\ell H = r$. □

6 The tail from k = 38001

Use the CRT construction already present in the development. Explicitly prescribe, for every prime $p \leq q$,

$$a_p \equiv q^- \pmod{p}\ (p < q^-), \qquad a_{q^-} = q^- - 1, \qquad a_q = q^- + 1.$$

Every $1 \leq m \leq 2q^- - 1$ belongs to one of these residues: away from $q^- - 1, q^-, q^- + 1$, use a prime divisor of $|m - q^-|$; at those three exceptions use $q^-, 2, q$, respectively. Choose $0 \leq a < P$ satisfying $a \equiv -a_p \pmod{p}$. Every $a + 2P + m$ is composite, giving a block inside $(2P, 4P)$ and a surrounding prime gap $g_- \geq 2q^- > t$.

The first integer of the block is $a + 2P + 1$ and is composite. Bertrand's theorem puts its preceding prime above half this cutoff, hence strictly above P. Thus the prefix at the descent contains every prime through P. Equations (S), (T), (D) give $\mu(t) > r$ there. Lemma 1 gives $R_r < t < g_- + 1$.

Let $M = \log(400P)$ and let N count primes in $(4P, 400P]$. Since $L \geq \ell \geq 3670$, one has $P \geq 2^{30}$. The bounds in Lemma 2 give

$$NM \geq (0.92 \cdot 400 - 1.12 \cdot 4)P = 363.52P.$$

Also $M < P/5$: use $\log P \leq 0.007P$ for $P \geq 1000$, $\log 400 < 6$, and $P \geq 2^{30}$. Put $G = \lceil 1.09M \rceil$. Then

$$(N-1)G \geq 1.09(N-1)M \geq 396.2368P - 1.09M > 396P,$$

because $0.2368 \cdot 5 > 1.09$. The finite average-gap principle now supplies a globally consecutive gap in this interval with

$$g_+ < G < 1.09M + 1, \qquad g_+ + 1 < 1.09M + 2.$$

Moreover $M = L + \log 400 < M_*$. The function $\log s + 1.1 + 4/s$ increases for $s > 4$. Lemma 4 and (A) imply $(g_+ + 1)A(400P) < r$, so $R_r > g_+ + 1$ at this later gap. Its coefficient denominator is positive: the earlier mean bound already supplies more than r primes.

The descent index is strictly below $\pi(4P)$, whereas the later gap starts at an index at least $\pi(4P)$. This establishes the required ordering even if the large gap's right endpoint is the first prime above $4P$. Hence every integer $k \geq 38001$ gives non-unimodality.

7 The existing finite range, without a prime-counting premise

The fixed certified gap supplies an earlier gap of length at least 4752, with preceding prime prefix containing every prime through

$$Y_* = \lfloor (\text{recordGapCenter} - 86399)/2 \rfloor, \qquad \log\log Y_* > 10.6.$$

The existing later twin-prime certificate supplies a pair $Q, Q+2$ with $\log(Q-1) > 9$ and $\log\log(Q-1) < 12.1$. Their ordering and these size inequalities are already proved in the source.

Apply (S) with $X = Y_*$, $t = 4752$. Replace $\log(1 + 1000/t)$ by the larger $\log 2 < 0.7$. Since $8.46 < \log 4752 < 8.47$ and $\log 8.47 < 2.137$,

$$\mu(4752) > 4752(10.6 - 2.137 - 0.7/8.46 - 2.2/10.46) > 4752(8.16) = 38776.32 > 37999.$$

Thus every $49 \leq k \leq 38000$, with $r = k - 1 \leq 37999$, has a strict descent there by Lemma 1. For the later twin, Lemma 4 gives

$$3A(Q-1) < 3(12.1 + 1.1 + 4/9) < 48 \leq r.$$

The lower coefficient comparison forces a strict ascent across that twin. The mean bound itself ensures enough primes for $e_r > 0$; there is no remaining count below 500000 or other prime-count certificate to supply.

Theorem 8. *Combine the existing closed proofs for $1 \leq k \leq 48$ and the existing fixed gap and twin certificates with the preceding derivation. For every positive integer k, the density sequence is unimodal exactly when $k \leq 3$, with no Dusart or finite-height RH premise.*

Proof. The domains are $1 \leq k \leq 48$, $49 \leq k \leq 38000$, and $k \geq 38001$. They are exhaustive and adjacent. The positive cases $1, 2, 3$ and the remaining cases through 48 are already proved in the existing source. □

8 Computational and formal status

Dependency	Treatment
Finite-height zeta zeros and completeness	Removed
Dusart error ray and sharp prime-counting bounds	Removed
New large finite verification interval	None
Original gap, twin primes, and cases through 48	Retained unchanged
Fixed scalar constants	Proved by rational inequalities and logarithmic tangents
New analytic estimates, tail, and all-k theorem	Implemented and compiled in Lean 4.34.0

The final declaration is `PrimeFactorUnimodality.completeClassification_elementary`. The entry point `Solution.lean` exports the same universal statement as the Mathlib-only challenge. The key new declarations are:

- `smoothedPrimeSum_lower`;
- `classification_descent_scalar`;
- `classification_ascent_scalar`;
- `elementary_tail_not_isUnimodal`.

All fixed logarithmic constants in the implementation follow from the certified bounds for $\log 2, \log 3, \log 5$ and the two tangent inequalities

$$\log b + 1 - b/x \le \log x \le \log b + x/b - 1 \qquad (x, b > 0).$$

Here b is an explicit rational product of powers of $2, 3, 5$. The infinite scalar ray is proved by differentiation, with no scan of integer parameters. There is no new prime or zero enumeration and no implicit finite cutoff to verify. Reusing the byte-verified release objects for the original fixed certificates affects build time only; the new mathematical proof replaces the analytic verification premise.

References

[1] S. Cambie, Resolution of Erdős' problems about unimodularity, arXiv:2501.10333.

[2] S. Wang and D. Crapis, A Complete Answer to Erdős Problem 690, arXiv:2605.08542v1, Theorem 1.1 and Corollary 1.2. The formalization adapts the symbolic strategy and replaces the explicit analytic inputs.

[3] J. N. Darroch, On the distribution of the number of successes in independent trials, Ann. Math. Statist. 35 (1964), 1317–1321. DOI.

[4] A. Gnedin, Cross Modality of the Extended Binomial Sums, Section 5.1, arXiv:2408.06477.

[5] S. Broadbent, H. Kadiri, A. Lumley, N. Ng, K. Wilk, Sharper bounds for the Chebyshev function $\theta(x)$, Section 1.1, arXiv:2002.11068. Only its historical elementary inequalities are used, not its later zero-based estimates.

[6] M. B. Villarino, Mertens' Proof of Mertens' Theorem, arXiv:math/0504289.

[7] Submission 4526 and its linked source and existing certificate proofs.